Presented by YEAAAH

Future City Intelligence.

Sensors and Smart Cities – Hack the Environment.

Agenda

1. The Challenge
2. Data and City Health Score
3. Smart City Control Hub
4. City Planner Dashboard
5. Citizen View
6. Use-cases, Impact and Future scope

Challenge

- Cities collect large volumes of air, weather & noise data
- Raw numbers are hard to interpret and rarely guide action
- Patterns, anomalies, and risks often go unnoticed
- Citizens don’t understand complex environmental indicators
- Need: a unified, intelligent system that explains what’s happening and what to do

Our Data Sources

- Multi-year datasets from Heilbronn:

    Air Quality: NO_2, PM10, PM2.5, O_3

    Weather: temperature, humidity, wind speed, rainfall

    Noise: day & night sound levels, disturbances

- All streams cleaned, merged, time-aligned and analyzed
- Powering dynamic dashboards for 3 user roles

Unified Health Score

- A single metric (0–100) summarizing overall air quality
- Integrates multiple pollutants into one understandable score
- Makes trends easier to compare across days & neighborhoods
- Enables quick interpretation for non-experts
- Forms the basis for alerts, risk detection, and city mood

Smart City Control Hub

- Designed for real-time situational awareness
- Air-Risk Episode detection (Health Score < 40 + high noise over time)
- Night Noise Events detected automatically between 22:00–06:00
- Hotspot mapping for faster on-site response
- Shows what is happening and where immediate action is needed

City Planner Dashboard

- Focuses on long-term city improvement
- Tracks chronic air & noise stress across the dataset
- Tree Priority Index identifies high-impact areas for greening
- Sensor Relationship Explorer shows correlations (e.g., NO_2 + noise = traffic)
- Noise Time-Travel Map simulates conditions at different hours
- Supports data-driven planning & policy design

Citizen View

- Simplifies complex data for everyday citizens
- Displays a City Mood emoji with qualitative explanation
- Shows the day's Health Score, noise levels & trends
- Clear, friendly language instead of technical metrics
- Helps residents understand when conditions are good or harmful

Delivered Use Cases

- Air Quality & Traffic Alerts for controllers
- Night-time Noise Monitoring for enforcement & event planning
- Green Infrastructure Planning using Tree Priority Index
- Emission Anomaly Detection through sensor correlations
- Scenario Planning with predictive noise mapping
- Supports mobility, environment, and public wellbeing goals
- Real life API example

Impact

- Faster responses to environmental hazards
- More informed, data-supported city planning
- Increased transparency and trust with citizens
- Identifies where interventions have the greatest effect
- Helps build a cleaner, quieter, healthier city

Future Scope

- Add heatwave + ozone early-warning system
- Connect to real-time IoT sensors
- Provide push alerts for citizens and city control
- Add more predictive models for emission sources
- Expand to more neighborhoods and sensors

40%

Faster identification of environmental risk periods

We’re excited to demonstrate our platform
Happy to answer questions

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